

CSE499A – Section 15 – Group 03

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CityPulse: Intelligent Parking & Traffic Control Platform

Weekly Progress Report – Update 1 (Sprint 1: Literature Review & Problem Analysis)

I. SPRINT OVERVIEW

Sprint 1 covers literature review and problem analysis. The goal is to survey related work, identify research gaps specific to the Bangladesh context, select datasets, and document findings before writing any implementation code. No build tasks are assigned to this sprint.

II. PAPERS REVIEWED

A. Real-time Bangla LPR on Low-Resource Hardware

Ashrafee et al. [1] propose a two-stage pipeline combining a Haar-cascade wake-up filter and MobileNet SSDv2 for Bangla license plate recognition. The system achieves 82.7% video detection rate and 27.2 FPS on a CPU (Intel i5-7200U) without any GPU. OCR F1 is 60.8%. The paper also releases the first public Bangla license plate video dataset (79 clips, 98 unique plates). A Flask integration is provided for immediate deployment.

B. Robust Bangla LPR with YOLOv8 and Vision-Language OCR

Hasin et al. [2] present the current state-of-the-art for Bangla ALPR. A two-stage YOLOv8m training strategy with phase-aware augmentation and progressive layer unfreezing achieves 97.83% localization accuracy. The OCR stage pairs a ViT-Base encoder with a BanglaBERT decoder, reaching 13.23% CER – a substantial improvement over the 2021 baseline. Training covers 13,629 images across all 64 districts of Bangladesh and is validated on external toll booth CCTV footage.

C. Intelligent Traffic Signaling in Dhaka

Azam et al. [3] deploy YOLO-based vehicle detection on the NHT-1071 dataset (1,071 images, 30,000 labeled objects, 4 classes for non-lane-based heterogeneous traffic) and apply NSGA-II multi-objective optimization to minimize total congestion and red-signal waiting time simultaneously. Field-tested on a Raspberry Pi 4B at Palashi Intersection, Dhaka. The NSGA-II framework is transferable to multi-lot parking slot allocation.

D. DNCC Smart Parking App, Gulshan

The Dhaka North City Corporation launched a production parking system in November 2023 managing 202 on-street parking spots across 8 roads in Gulshan [4]. Features include real-time availability display, 12-hour advance reservation, bKash/Nagad payments, and a 98% complaint resolution rate via the Dhaka app. The system depends entirely on manual booth operators with no computer vision component, which defines the opportunity CityPulse addresses.

E. Real-time Parking Occupancy Detection (NSU)

Prova et al. [5] from North South University apply miniAlexNet to PKLot and CNRPark+EXT, achieving 96.1% on PKLot and 90.1% on CNRPark in cross-dataset evaluation. On locally collected Dhaka apartment CCTV footage they reach 86.1% indoor and 90.5% outdoor accuracy without any Bangladesh-specific training data. This establishes the minimum performance baseline and confirms that transfer learning from international benchmark datasets to Bangladeshi conditions is viable.

III. DATASET ANALYSIS

PKLot [6]: 695,899 pre-cropped 150×150 patches from three parking lots across sunny, overcast, and rainy conditions. Verified patch count after download: 695,851 (48 files excluded, <0.01%). Class distribution: 337,780 occupied, 358,071 empty. A lot-based split is applied to prevent data leakage from near-duplicate frames captured 5 minutes apart on the same day.

TABLE I
PKLOT PATCH COUNTS PER LOT AND WEATHER CONDITION

Lot	Weather	Days	Occ.	Empty
PUC	Cloudy	11	42,363	90,417
	Rainy	8	55,105	27,951
	Sunny	25	96,761	111,626
UFPR04	Cloudy	15	11,608	27,777
	Rainy	14	2,351	5,607
	Sunny	20	32,166	26,334
UFPR05	Cloudy	19	33,764	23,202
	Rainy	8	6,078	2,851
	Sunny	25	57,584	42,306
Total			337,780	358,071

Split strategy: PUC + UFPR04 assigned to training (530,066 patches), UFPR05 assigned to validation (165,785 patches). UFPR05 provides an unseen camera angle for honest evaluation.

CNRPark+EXT: 113,140 labeled 150×150 patches from 9 cameras across overcast, rainy, and sunny conditions. Official author-defined splits are: train.txt (94,493 entries, 47,616 occupied + 46,877 empty), val.txt (18,647 entries, 13,415 occupied + 5,232 empty). Zero missing files confirmed.

Combined prepared dataset: 808,991 total patches published at <https://kaggle.com/datasets/raahad/parking-occupancy-merged>.

TABLE II
FINAL MERGED DATASET COUNTS

Split	Occupied	Empty
train	287,970	336,589
val	110,841	73,591
Total	398,811	410,180

Bangladesh gap: No public parking lot occupancy dataset exists for Bangladesh after exhaustive search across Kaggle,

Hugging Face, arXiv, and Mendeley. Local data collection in Sprint 4 is required for fine-tuning.

IV. LEGAL & REGULATORY REVIEW

Bangladesh has no dedicated CCTV surveillance law or code of practice. The Personal Data Protection Ordinance (2025), enforced from May 2027, will classify license plate data as personal data since plates link to vehicle owners through BRTA records. Practical compliance requirements after enforcement are entrance signage stating that cameras and LPR are in use, a data retention policy limiting storage to billing and dispute resolution purposes, and a restriction on third-party data sharing without consent. No regulatory barrier exists to operating a parking management system before enforcement begins in 2027.

V. KEY FINDINGS

The NSU Dhaka paper result – 86% accuracy on Bangladeshi CCTV footage without local training data – establishes the transfer learning approach as viable and sets the performance floor for Sprint 4. The DNCC deployment validates market acceptance of app-based parking in Dhaka and the existing bKash/Nagad payment infrastructure can be integrated directly. No public Bangladeshi parking dataset exists, confirming the need for original data collection.

VI. PLAN FOR SPRINT 2

System architecture finalization, database schema design (PostgreSQL + MongoDB), API contract specification, UI/UX wireframes for all three client interfaces (driver mobile app, owner portal, authority dashboard), and Docker Compose environment setup with empty service shells.

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